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# CS516: Cloud Computing

Thao Huy Vu, M.S.

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# Course Overview

| | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|---|---|---|---|---|---|---|
| Theme I: Life is structured in layers | | | | | | |
| **Week 1** | AWS Introduction | IAM | IAM | Compute Offerings | VPC & Routing | Storage |
| Theme II: The Nature of Life is to Grow | | | | | | |
| **Week 2** | SQL Database Offerings | Midterm Review | Midterm Exam (10:00 AM – 12:30 PM) | Advanced Compute Offerings | NoSQL Database Offerings | Application Services |
| Theme III: Purification Leads to Progress | | | | | | |
| **Week 3** | Workshop 1: Scale App with Servers | Infrastructure as Code (IaC) and CI/CD | Workshop 2: Full-stack App with Serverless | AWS Well-Architected Framework | Final Review & Project | Project |
| Theme IV: Integrating all parts together - The Whole is Greater than the Sum of the Parts | | | | | | |
| **Week 4** | Project | Project Presentation | Project Presentation | Final Exam (10:00 AM – 12:30 PM) | | |

# CS516
# Cloud Computing
**The evolution of the individual and the evolution of the environment are naturally interdependent**

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# COURSE GOAL

The goal of this course is to equip students with a comprehensive understanding of Cloud Computing and practical skills in using Amazon Web Services (AWS). Upon completing this course, students can architect and deploy a full-stack application with security, scalability, and cost-efficiency.

# Course Objectives

By the end of the course, students will be able to:

- **Architect and Deploy Applications**: Learn to design, deploy, and manage scalable, highly available, and fault-tolerant systems on AWS.
- **Understand and Navigate AWS**: Gain a thorough understanding of the AWS ecosystem, including core services like Compute, Storage, and Networking.
- **Implement Security Measures**: Master the use of AWS Identity and Access Management (IAM) to secure applications and data.
- **Utilize Data Services**: Develop skills to integrate various AWS database services to store and manage data efficiently.
- **Leverage Advanced AWS Services**: Explore and apply advanced services such as SNS, SQS, Server-less, Scalability to enhance the functionality and capability of applications.

# References

1. AWS: https://docs.aws.amazon.com
2. ChatGPT: https://chatgpt.com
3. Google AI: https://gemini.google.com

# Important Notes

Students are required to use regular AWS accounts to gain hands-on experience. Please be aware of the following:

**Billing responsibility**: Students are solely responsible for the charge incurred in their AWS accounts. Neither MIU nor the instructor are responsible for any costs associated with students' use of their AWS services.

**Cost management**: It is highly recommended that students leverage the AWS Free Tier, clean up resources after use, and regularly check their billing to manage and minimize costs.

**No Makeup Exam:** You should arrange your personal matters to ensure you can attend the exams. If you miss the exams, you get an NC for this course.

# Evaluation Criteria

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|---|---|
| Midterm Exam | 30% |
| Final Exam | 40% |
| Project | 20% |
| HW & WS | 10% |
| Professionalism Reduction | -5% to 0% |

## Grading scale

| | |
|---|---|
| A | 93-100 |
| A- | 90 - 92 |
| B+ | 87 - 89 |
| B | 83 - 86 |
| B- | 80 - 82 |
| C+ | 77 - 79 |
| C | 73 - 76 |
| C- | 70 - 72 |

NC 00 - 69

# Contact Info

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McLaughlin #209

# Class Attendance Requirements

1. **Class Hours:**
    a. Morning Session: 10:00 AM – 12:30 PM, Monday – Saturday
    b. Afternoon Session: 1:30 PM – 3:30 PM, Monday – Friday
2. **Attendance Obligation:**
    a. In-person students are required to attend all classes, except under exceptional circumstances.
    b. If you cannot attend a class, you must seek my approval by sending an explanatory email.
3. **Missed Classes:**
    a. Missing three classes without prior approval may warrant consideration of withdrawing from the course.
4. **Homework Submission:**
    a. Failure to submit your first homework solution three times before the deadline (10:00 PM) will result in the loss of all related homework grades.
5. **Punctuality:**
    a. Please arrive on time to prevent any disruptions for fellow students.
6. **Importance of Attendance:**
    a. Attendance is mandatory, as all class components (lectures, discussions, Q&A, labs) contribute to the learning process.
7. **Excused Absences:**
    a. Excused absences are typically granted for illness or family emergencies.

8. **Advance Notice:**
    a. If you anticipate missing a class, please communicate with me beforehand.
9. **First Lesson Importance:**
    a. Attendance is particularly crucial for the first lesson of each course. Students not present on the first morning (excluding valid reasons like illness or family emergencies) may be advised to withdraw.
10. **Unexcused Absences:**
    a. Unexcused absences may result in an NC (No Credit) grade for the entire class.

Your active participation and attendance are key to your successful learning experience. If you have any questions or concerns, please don't hesitate to reach out to me.